

AIMLZG546 - Software Engineering for Machine Learning

Assignment I - Report

Group No: 13

Project: Credit Card Fraud Detection

Group Details

Sl. No	BITS ID	Name	Contribution of team member (Qualitative)	Percentage Contribution (out of 100)
1	2025ae05576	Anirudh Anand	Report lead. Wrote the domain and problem statement, ML problem formulation, stakeholders, scope, functional requirements and measurable goals (Sections 1 and 2) and the GR4ML Business View. Compiled the final report and PDF.	25
2	2025ae05178	Aniketh Paul	Wrote the GR4ML Analytics Design View, the Data Preparation View and the top three quality requirements with justification (Sections 3.2, 3.3 and 4). Built the data part of the pipeline in the notebook: EDA, ingest, clean, features and split filters.	25
3	2025ae05898	Pushadapu Sanjay Kumar	Designed the system architecture diagram showing ML and non-ML components, selected the two patterns and wrote Sections 5 and 6. Built the FastAPI prediction service, input validation, prediction logging and the automated tests.	25
4	2025af05111	Biswajeet Mahato	Built the model part of the pipeline: train and evaluate filters, threshold selection on validation, MLflow tracking and model registry. Ran the notebook end to end, took the screenshots and wrote Sections 7.2, 7.3, 8 and 9.	25
			Total	100

Objective 1: Requirements Formulation

1. Domain and Problem Statement

1.1 Domain

The domain we chose is banking, and within it credit card fraud detection. Millions of card payments happen every day and each one has to be approved or declined in a few seconds. If a fraudulent payment goes through, the bank normally has to bear the loss and the customer also loses trust in the bank. So we felt this is a good practical area for a machine learning application.

1.2 Problem Statement

Banks lose a lot of money every year because of fraudulent credit card transactions. The fraud team cannot check every payment by hand, and manual checking is too slow to stop a payment that is being processed. Fraud is also very rare compared to genuine payments (less than 1%), so fixed rules either miss the fraud or block many genuine customers.

In this assignment we propose an application that checks each card transaction when it arrives and gives the probability that it is fraud. Transactions with a high probability are flagged, and the fraud analysts look only at the flagged ones instead of all transactions. Our goal is to catch at least 90% of the frauds, wrongly flag less than 2% of genuine transactions, and give the answer fast enough to be used during the payment.

1.3 Why Machine Learning is Needed

Most banks already have rule based checks, for example blocking a payment above some amount or from an unusual country. Someone has to write these rules by hand, and fraudsters slowly find ways around them. A rule also looks at only one or two fields at a time, and if the rules are made stricter, more genuine payments get declined.

A machine learning model can learn the fraud patterns from old transactions that are already marked as fraud or genuine. It can look at many features together and can be trained again when new fraud patterns come. The model also gives a probability instead of a plain yes or no, so the bank can pick a threshold that gives an acceptable balance between missed frauds and false alarms.

1.4 Stakeholders

Stakeholder	Interest in the system
Bank risk manager	Reduce fraud losses without increasing complaints from customers
Fraud analyst	Get a reliable score so that only suspicious transactions need manual review
Card holder (customer)	Genuine payments should not be declined
ML and development team	Build and maintain a model and service that are accurate and fast

1.5 Scope

These are inside the scope of this assignment:

- Checking a single transaction and returning a fraud label with its probability
- Providing this check through a prediction service (API)
- Training the model and storing it in a model registry
- Logging every prediction made by the service

Outside the scope are: connecting to a real payment network, sending alerts to customers, handling chargebacks and using real customer card data. We use a public anonymised dataset instead.

1.6 Formulation as a Machine Learning Problem

We treat this as a binary classification problem. Each transaction is labelled fraud (1) or genuine (0). The model takes the transaction features as input and gives the probability of fraud, and the final label

is decided by a threshold.

We use the "Credit Card Fraud Detection" dataset from Kaggle. It has card transactions made by European card holders, with 284,807 transactions of which only 492 (0.173%) are fraud. Except for Time and Amount, the features V1 to V28 are already anonymised by the dataset provider. The two classes are very unbalanced, so accuracy alone does not tell us much (a model that says everything is genuine would still be 99.8% accurate). Because of this we judge the model with recall, precision, F1 score, the area under the precision-recall curve and the confusion matrix.

For the design we picked two architectural patterns: Pipe-and-Filter for the training pipeline, and Microservices for serving the model as a separate prediction service together with a model registry. They are explained in Section 6.

2. Requirement Specifications and Measurable Goals

2.1 Functional requirements

ID	Requirement
FR1	The system accepts one card transaction (Time, V1 to V28 and Amount) through a POST /predict API call.
FR2	The system returns the fraud probability, the fraud label, the threshold used and the model version.
FR3	The system rejects invalid input (missing field, wrong type, NaN or infinity, negative Amount, unknown field) with HTTP 422 and does not return a prediction for it.
FR4	The system writes one row to a prediction log for every prediction made.
FR5	The training pipeline reads the dataset, cleans it, splits it, trains the model and reports the evaluation metrics. The pipeline can be run again to retrain.
FR6	The trained model is stored in a model registry, and the prediction service loads only the registered model marked as champion.
FR7	The service has a health endpoint that reports whether it is running and which model version it serves.

2.2 Measurable goals

Goal	GR4ML concept	Metric	Target	Result (latest run)	Met?
Catch fraud	Indicator of the strategic goal; softgoal High recall	Recall on the test set	at least 90%	85.3%	No
Avoid blocking genuine customers	Indicator of the strategic goal; softgoal Few false alarms	False-positive rate on the test set	below 2%	1.08%	Yes
Decide while the payment is processed	Softgoal Low latency	p95 latency of the prediction API	below 200 ms	5.9 ms	Yes
Reduce losses	Strategic goal	Share of fraud money caught (test set)	at least 30%	82.4%	Yes

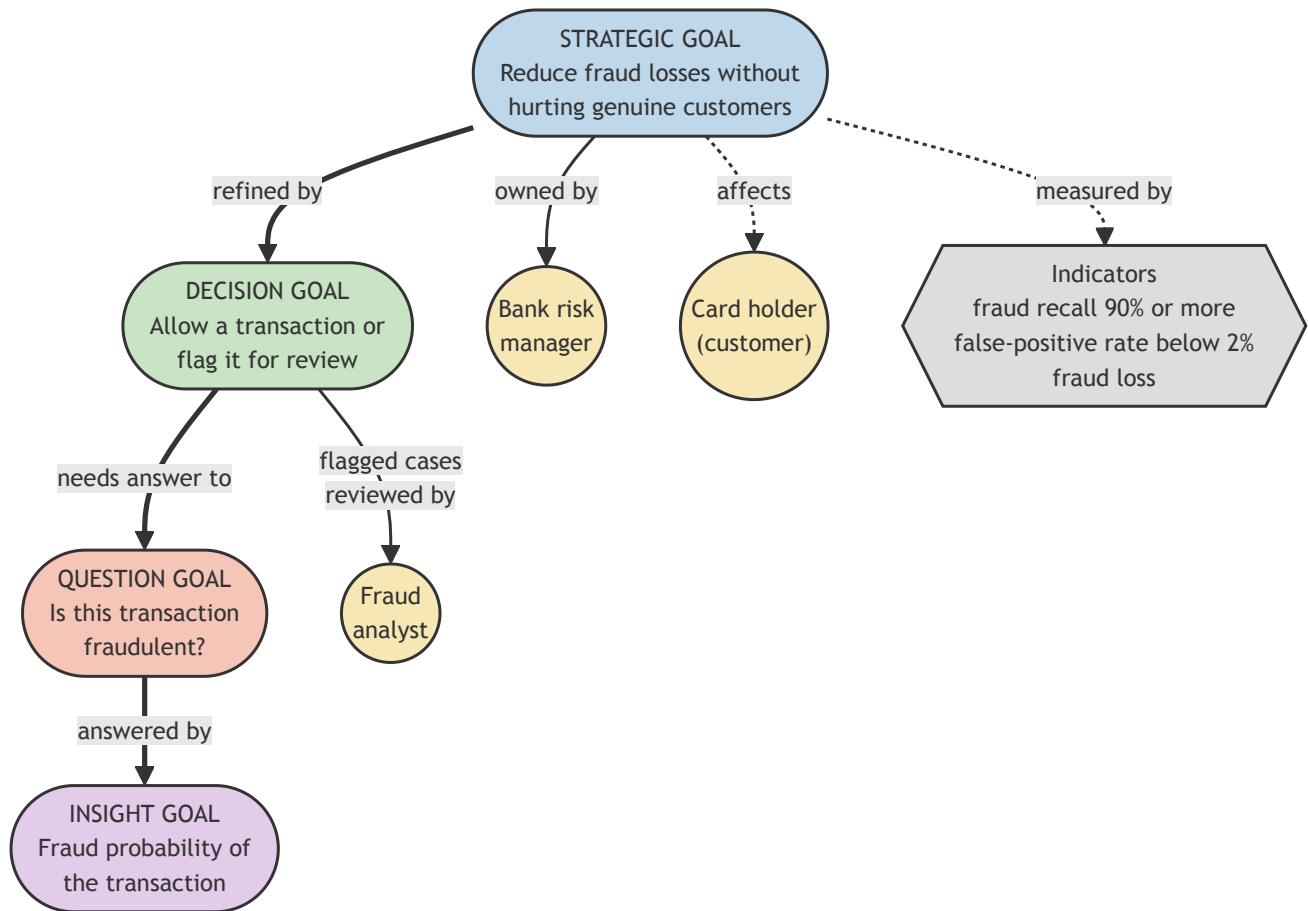
The results are from our latest notebook run. If a target is not met, we report the real value and do not tune anything on the test set.

2.3 Business rules

ID	Rule
BR-001	A transaction is labelled fraud only if its fraud probability is greater than or equal to the threshold.
BR-002	The threshold is chosen on the validation set (highest value with recall at least 0.90), stored as a run parameter and kept in one config place.
BR-003	The scaler is fitted on the training data only.
BR-004	The scaling is saved inside the model pipeline and applied the same way at serving time.
BR-005	Evaluation reports recall, precision, F1, PR-AUC, ROC-AUC, false-positive rate and the confusion matrix.
BR-006	Targets are compared honestly with the test results; nothing is tuned on the test set.
BR-007	Invalid input is rejected with HTTP 422 and is not logged.
BR-008	One log row is written per prediction; a failed write is reported as an error.
BR-009	The API loads only the model with the alias champion.
BR-010	Training runs as a chain of six filters: ingest, clean, features, split, train, evaluate.
BR-011	A fixed seed (42) and a single config make the run reproducible.

3. GR4ML Views

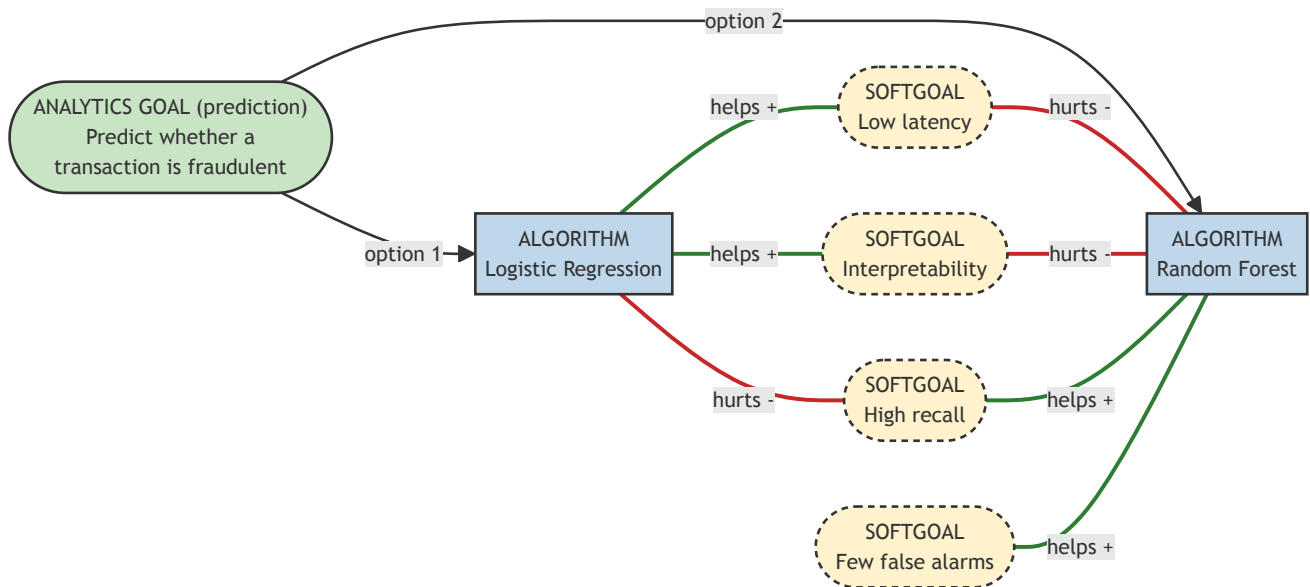
3.1 Business View



Colour key: yellow circles are actors, the blue, green, red and purple rounded boxes are the strategic, decision, question and insight goals, and the grey hexagon holds the indicators. Thick arrows connect the goals, dotted lines are weaker links (affects, measured by).

Explanation: The bank risk manager owns the strategic goal, which is to reduce fraud losses without hurting genuine customers. The card holder is affected because a wrongly blocked payment is annoying for them. To reach the goal, the bank must decide for every transaction whether to allow it or to send it to a fraud analyst for review. For this decision the bank needs to know one thing, whether the transaction is fraudulent, and the insight that answers it is the fraud probability of the transaction. We measure the strategic goal with three indicators: fraud recall, false-positive rate and fraud loss. The first two are the same measurable goals as in Section 2.2.

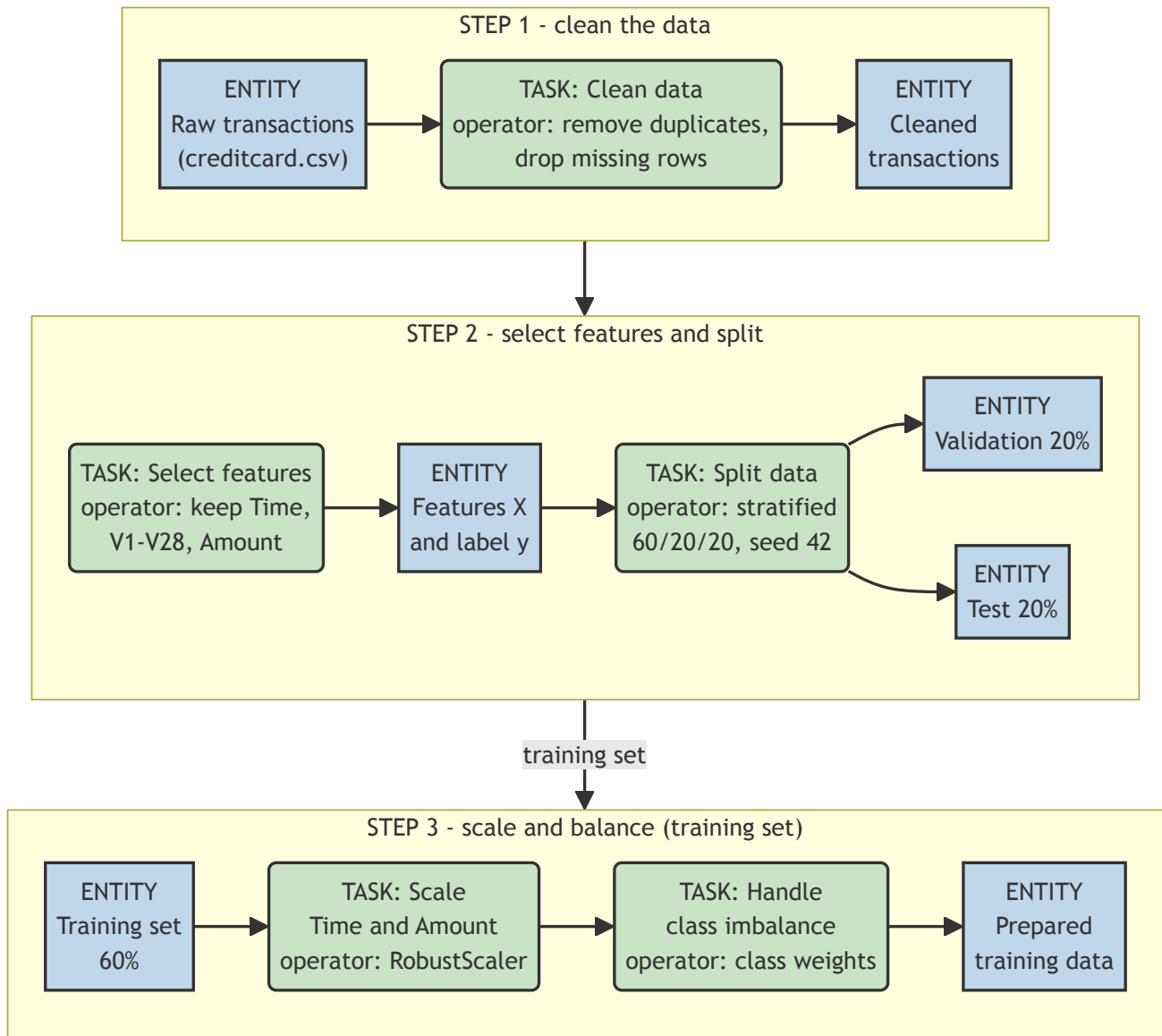
3.2 Analytics Design View



Colour key: green arrows are positive influences (+) and red arrows are negative influences (-). These are expected influences, not measured results.

Explanation: The analytics goal is prediction, we want to predict whether a transaction is fraud. We considered two algorithms for this, Logistic Regression and Random Forest, and implemented both. The softgoals are high recall, low latency, interpretability and few false alarms. We expected Logistic Regression to be good for latency and interpretability but weaker on recall, and Random Forest to be better on recall and false alarms but worse on interpretability and latency. These influence links are only our expectations from the course material, not results. In our run Logistic Regression was selected because it had the lower validation false-positive rate (1.02% against 4.69% for Random Forest), although Random Forest had the better PR-AUC (0.859 against 0.786). The recall of Logistic Regression was 90.4% on validation and 85.3% on the test set.

3.3 Data Preparation View



Colour key: blue boxes are entities (data), green boxes are tasks, and the operator used by a task is written inside the task box. Arrows show the flow of the data.

Explanation: First the raw transactions are cleaned by dropping duplicate rows and rows with missing values. From the cleaned data we make the feature table, with Time, V1 to V28 and Amount as inputs and Class as the label. This table is split into training, validation and test sets (60/20/20, stratified, seed 42). Time and Amount are scaled with a RobustScaler that is fitted on the training set only, and the class imbalance is handled with class weights in the classifier. We use the validation set to choose the decision threshold and keep the test set only for the final result.

3.4 GR4ML concept mapping

GR4ML concept	In this project
Strategic goal	Reduce losses from card fraud
Decision goal	Block or allow a transaction
Question	Is this transaction fraudulent?

GR4ML concept	In this project
Insight	Fraud probability and label per transaction
Indicators	Recall, false-positive rate, p95 latency
Analytics goal	Binary classification (fraud or genuine)
Algorithms	Logistic Regression, Random Forest
Softgoals	High recall, few false alarms, low latency
Data preparation entities	Raw transactions, cleaned data, features, train/validation/test sets
Tasks and operators	Ingest, clean, features, split, train, evaluate

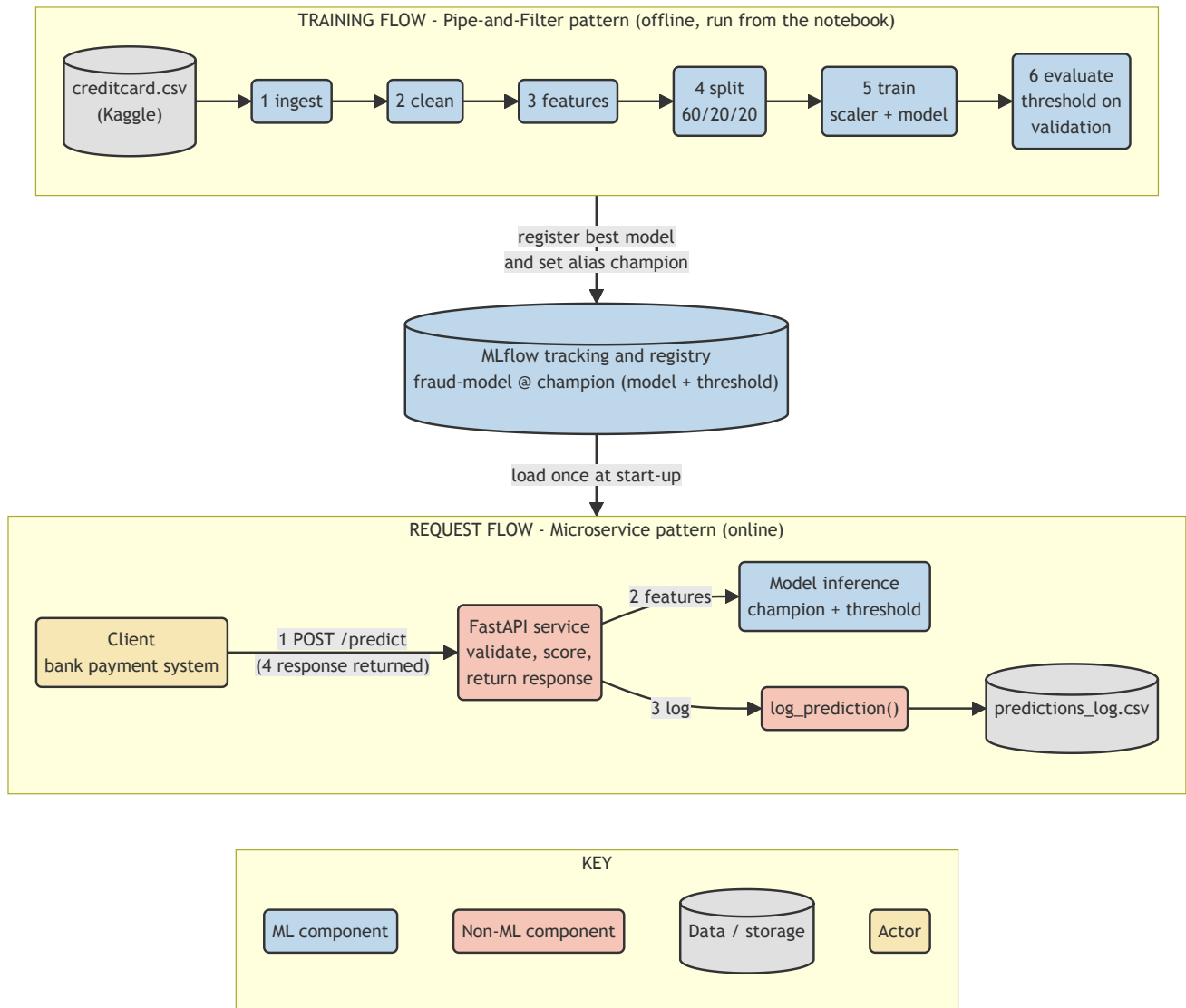
4. Top Three Quality Requirements

#	Quality requirement	Measurable target	Justification (why it matters here)	How architecture supports it
1	Accuracy (recall)	Recall at least 90% with false-positive rate below 2%	A fraud that is not detected is a direct loss for the bank. The classes are very unbalanced, so overall accuracy would hide the missed frauds.	We use class weights and a separate validation set to choose the threshold, and we report recall, precision, F1, PR-AUC and the confusion matrix instead of accuracy.
2	Performance (latency)	p95 latency below 200 ms	The decision has to be taken while the payment is being processed, so a late answer is as bad as no answer.	The model is loaded once when the service starts and kept in memory. The prediction service is separate from training, so for each request it only validates the input and calls the model once.
3	Security and privacy	No card number accepted or stored; invalid input rejected	Card data is sensitive, and a service that accepts any input can be misused.	The API accepts only the numeric model features (no card number or name) and rejects any extra field. The prediction log stores only the probability, label, latency and model version.

Authentication of API clients is not part of this assignment. We list it as a limitation of our prototype.

Objective 2: System Architecture

5. System Architecture Diagram



The key inside the diagram explains the colours: blue is an ML component, red is a non-ML component, grey is data or storage and yellow is an actor.

ML components: the six pipeline filters (ingest, clean, features, split, train, evaluate), the MLflow model registry that keeps the trained model and its threshold, and the model inference step inside the prediction service.

Non-ML components: the FastAPI service with the /predict and /health endpoints, the pydantic request validation, the log_prediction function with the prediction log file, and the client that sends the transactions.

Request flow: (1) The client sends a transaction to /predict. (2) The FastAPI service validates it and gives the features to the loaded champion model, which returns a probability. (3) The service compares the probability with the threshold and writes one row to the prediction log through log_prediction(), and (4) the response goes back to the client.

Training flow: the Kaggle dataset goes through the six filters. In the last filter the threshold is chosen on the validation set, the results are logged to MLflow, and the best model is registered as fraud-model with the alias champion. The prediction service loads this model when it starts.

6. Architectural Patterns

6.1 Pattern 1: Pipe-and-Filter

- Problem: Training a model has several processing steps. If everything is in one long script, it is hard to test a single step or to replace one, for example the model.
- Solution: We divide the processing into independent filters. Each filter takes an input, gives an output and keeps no hidden state, and the output of one filter is the input of the next one.
- How we applied it: The training pipeline has six filters (ingest, clean, features, split, train, evaluate), each in its own module under pipeline/. The file run_pipeline.py only connects them. The train filter accepts a classifier name, so the other filters need no change, and we used this to train both Logistic Regression and Random Forest.
- Benefits and trade-offs: Each filter can be tested alone (the notebook has tests for this) and the order of the steps is easy to follow. The trade-off is that the data is passed between filters as whole tables in memory. That is fine for this dataset, but a much bigger dataset would need streaming.

6.2 Pattern 2: Microservices

- Problem: The payment system of the bank needs the model at any time, while training is a separate job that runs only sometimes and needs far more resources.
- Solution: We make the prediction a small independent service with its own interface and keep it apart from the training code. Managing the models is done by a third component, the registry.
- How we applied it: The FastAPI service in api/main.py only loads the registered model fraud-model with the alias champion and answers /predict and /health. It never trains. MLflow works as the registry between training and serving, so a new model can be registered and picked up without changing the service code.
- Benefits and trade-offs: The service can be restarted or scaled without touching the training, and every response carries the model version. The trade-off is that there are more parts (service, registry and log) than in a single script, and the service has to be restarted to load a newly registered champion.

7. Implementation

7.1 Technology stack

Layer	Technology	Purpose
ML pipeline	Python, pandas, scikit-learn	Filters for data preparation, training and evaluation
Model registry	MLflow (SQLite backend)	Tracking of runs and registry of the champion model
Prediction service	FastAPI, pydantic, uvicorn	Separate prediction microservice with input validation

Layer	Technology	Purpose
Logging	CSV file written by log_prediction()	One row per prediction
Testing	pytest, FastAPI TestClient	Tests named after the business rules
Runtime	Jupyter notebook 13.ipynb (runs in Google Colab or VS Code)	Runs everything from top to bottom, no GPU needed

7.2 Pipe-and-Filter pipeline

The notebook writes each filter to its own file and run_pipeline.py joins them. ingest checks that the file and the columns exist, and stops with an error if they do not. clean drops duplicate rows and rows with missing values. features takes Time, V1 to V28 and Amount as inputs and Class as the label. split does a stratified 60/20/20 split with seed 42. train fits a scikit-learn Pipeline where a RobustScaler for Time and Amount comes before the classifier, so the scaler is fitted on the training set only and is saved together with the model. evaluate picks the highest threshold that gives a recall of at least 0.90 on the validation set, and the test set is used only once at the end with this threshold.

```
# each stage hands its output to the next one
raw = ingest(data_path)
df = clean(raw)
X, y = build_features(df)
parts = split(X, y)
# train every candidate on the training set and judge it on the validation set
fitted = {}
for name in candidates:
    model = train(parts["X_train"], parts["y_train"], name)
    threshold, val_m = evaluate_validation(model, parts["X_val"], parts["y_val"])
    fitted[name] = (model, threshold, val_m)
# the winner is chosen from validation results only; the test set is used once at the end
winner = pick_winner(fitted) # lowest validation false-positive rate, PR-AUC breaks ties
model, threshold, val_m = fitted[winner]
test_m = evaluate_test(model, parts["X_test"], parts["y_test"], threshold)
```

Model results: The dataset has 284,807 rows with 492 frauds (0.173%). After removing 1,081 duplicate rows, 283,726 rows remain with 473 frauds (0.167%). The stratified split gives 170,235 training, 56,745 validation and 56,746 test rows. For each model the threshold is the highest one that still gives a validation recall of at least 0.90.

Model (validation)	Threshold	Recall	Precision	FPR	PR-AUC
Logistic Regression	0.7692	0.9043	0.1286	1.02%	0.786
Random Forest	0.0107	0.9043	0.0310	4.69%	0.859

Both models were tuned to the recall target on validation. Random Forest gives 4.69% false positives on validation, which is above the 2% limit, but Logistic Regression stays within it, so we selected Logistic Regression (version 1 in the registry). Below are the results of the selected model on the test set, which we used only once:

Recall	Precision	F1	PR-AUC	ROC-AUC	FPR
0.8526	0.1169	0.2056	0.6884	0.9644	1.08%

Confusion matrix on the test set: TN 56,039, FP 612, FN 14, TP 81. **The FPR target (below 2%) is met, but the recall target (at least 90%) is not met on the test set (85.3%).** On validation the recall was 90.4%. The test set has only 95 frauds and the validation set 94, so missing just one more fraud changes recall by about one percentage point, and a threshold that is only just enough on validation can fall short on new data. We report the result as it is and did not tune the threshold on the test set.

Fraud loss (goal G4): We measured the loss in money using the Amount column and assumed that every flagged fraud is stopped. The 95 frauds in the test set add up to 14,766.31 and the model catches 12,163.97 of it, so the fraud loss goes down by 82.4%, against the target of 30% (met). The cost is that genuine transactions worth 205,692.94 were also flagged, and a real bank would have to consider the cost of these blocked payments too. This is only an estimate from the test set and it does not include chargeback fees or the cost of reviewing the flagged transactions.

Note on the selection rule: In our first run we chose the winner by the best validation PR-AUC and Random Forest won. On the test set it missed both targets (recall 87.4%, FPR 4.74%). After seeing this we changed the rule to the lowest validation FPR, with PR-AUC as tie-break, because the false-alarm limit is one of our stated goals. The change was based on validation results only, but we made it after seeing the first run, so we mention it here.

7.3 MLflow model registry

Every candidate model is logged as an MLflow run with its parameters (model name, seed, threshold) and its validation metrics. The winning model also gets its test metrics, is registered as fraud-model, and the alias champion is set on this version. The threshold is saved as a run parameter, so the prediction service reads the same value that was chosen during evaluation and it is not written inside the service code. After the run, fraud-model has Logistic Regression as version 1 with the alias champion, and the stored threshold is 0.7692. The prediction service loads only `models:/fraud-model@champion`, so a new version can be promoted without any change in the service code.

7.4 Prediction microservice (FastAPI)

The service has two endpoints. GET /health returns the status and the model version. POST /predict takes the 30 input values and returns request_id, probability, is_fraud, threshold, model_version and latency_ms. The label is true only when the probability is greater than or equal to the threshold. Invalid input gets HTTP 422 and is not logged. If the prediction log cannot be written, the call fails with HTTP 500 and does not continue silently.

```

@app.post("/predict", response_model=Prediction)
def predict(tx: Transaction) -> Prediction:
    """Score one transaction; the label is probability >= threshold and nothing else (BR-001)."""
    start = time.perf_counter()
    # Invalid requests never reach this point, FastAPI has already answered with 422.
    row = pd.DataFrame([tx.model_dump()], columns=config.FEATURES)
    # The saved model scales the input itself, so no separate preprocessing is done here (BR-004).
    probability = float(app.state.model.predict_proba(row)[0, 1])
    is_fraud = probability >= app.state.threshold
    latency_ms = (time.perf_counter() - start) * 1000
    request_id = str(uuid.uuid4()) # unique id so a log row can be matched to a call
    try:
        log_prediction(request_id, probability, is_fraud, latency_ms, app.state.version)
    except OSError as exc:
        # A failed log write is reported as an error, never ignored (BR-008).
        raise HTTPException(status_code=500, detail=f"Prediction log failed: {exc}") from exc
    return Prediction(request_id=request_id, probability=probability, is_fraud=is_fraud,
                      threshold=app.state.threshold, model_version=app.state.version,
                      latency_ms=round(latency_ms, 3))

```

Measured latency: For 100 requests the median is 4.4 ms, the p95 is 5.9 ms and the maximum is 9.3 ms. The target of p95 below 200 ms is met.

7.5 Prediction logging

Every successful prediction is added to `predictions_log.csv` with the time, request id, probability, label, latency and model version. We do not write the feature values to the log, so no transaction data is stored there.

```

def log_prediction(request_id: str, probability: float, is_fraud: bool, latency_ms: float,
                  model_version: str, path: Path | None = None) -> None:
    """Write one log row; an OSError is raised if the file cannot be written."""
    path = Path(path or config.LOG_PATH)
    # Only result data is logged. The transaction values themselves are not stored.
    row = {"timestamp": datetime.now(timezone.utc).isoformat(), "request_id": request_id,
          "probability": round(probability, 6), "is_fraud": int(is_fraud),
          "latency_ms": round(latency_ms, 3), "model_version": model_version}
    with _lock:
        new_file = not path.exists() or path.stat().st_size == 0 # write the header once
        with path.open("a", newline="") as f: # "a" = append, old rows are kept
            writer = csv.DictWriter(f, fieldnames=FIELDS)
            if new_file:
                writer.writeheader()
            writer.writerow(row)

```

7.6 End-to-end run and test calls

We run the notebook from top to bottom after restarting the runtime. It uploads the dataset, runs the pipeline, shows the MLflow registry and starts the service in the background. Then it sends one fraud row and one genuine row from the test set, three invalid requests (negative Amount, missing field,

unknown field) and 100 requests to measure the latency. At the end it shows the prediction log and runs the automated tests with pytest. The last pytest run shows **31 passed**.

8. Application Screenshots

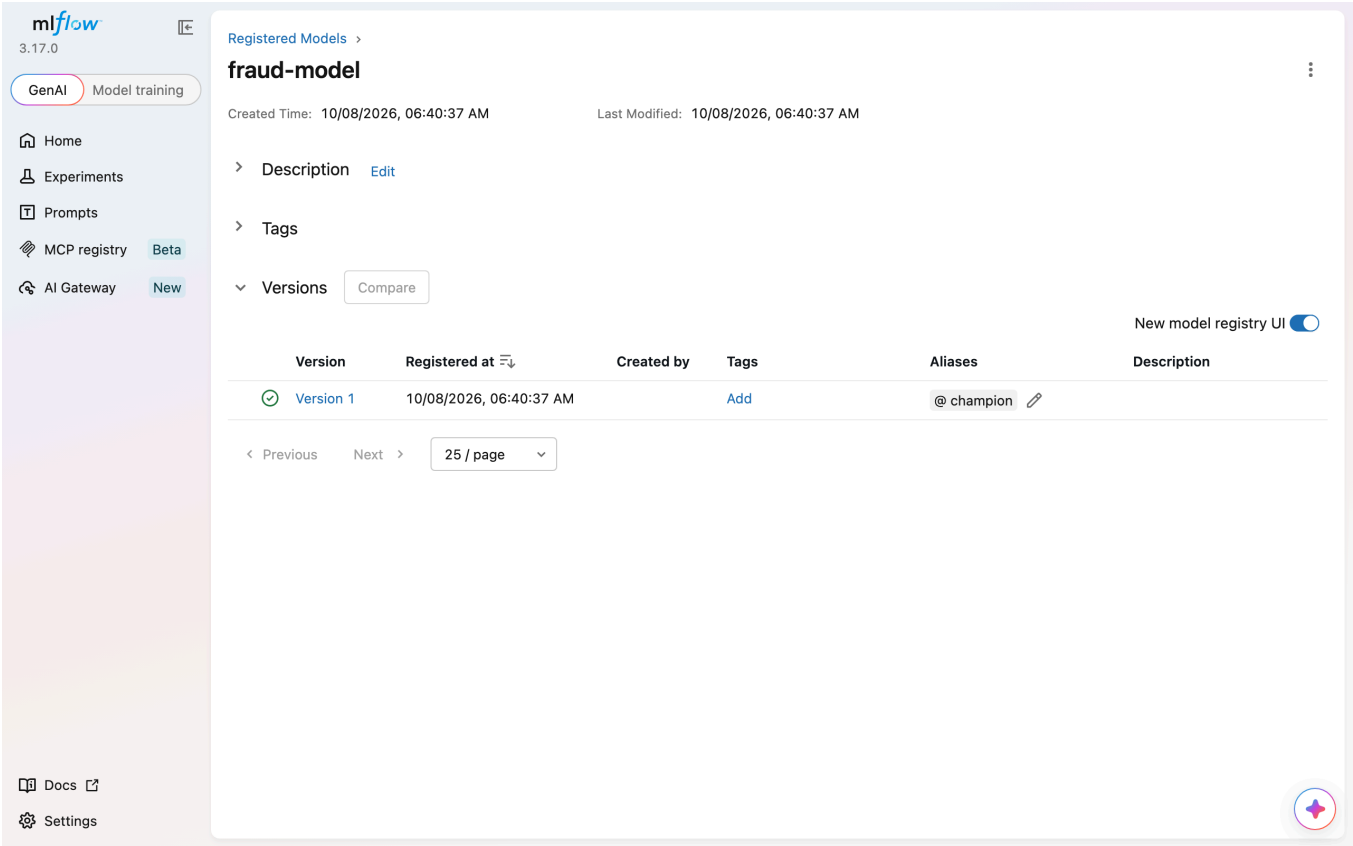
Screenshot 1 - Pipeline run. Output of the six-filter pipeline: 284,807 rows read, 1,081 duplicates removed, the 60/20/20 split, both models trained and Logistic Regression selected, with test recall 0.853 and FPR 0.0108.

Pipeline run (13.ipynb)

```
[1/6 ingest]  rows=284807 columns=31
[2/6 clean]   rows=283726 (removed 1081), fraud=473 (0.167%)
[3/6 features] X=(283726, 30)
[4/6 split]  train=170235 val=56745 test=56746
[5/6 train]  logistic_regression val recall=0.904 precision=0.129 pr_auc=0.7860 threshold=0.7692
[5/6 train]  random_forest      val recall=0.904 precision=0.031 pr_auc=0.8592 threshold=0.0107
[6/6 evaluate] winner=logistic_regression test recall=0.853 precision=0.117 f1=0.206 pr_auc=0.6884 fpr=0.0108
Targets on test set: recall>=0.90 -> False, fpr<0.02 -> True
Successfully registered model 'fraud-model'.
Created version '1' of model 'fraud-model'.
Registered fraud-model version 1 with alias 'champion'
```

Winner (lowest validation FPR): logistic_regression | model version: 1
Targets on the test set: {'recall_ok': False, 'fpr_ok': True, 'loss_ok': True} (recall >= 0.90, FPR < 0.02)

Screenshot 2 - MLflow. MLflow registry page of fraud-model. Version 1 is registered and has the alias champion, which is the only model the service loads.



Screenshot 3 - FastAPI. The /docs page of the service with the GET /health and POST /predict endpoints, the request fields (Time, V1 to V28, Amount) and the response fields.

Credit Card Fraud Detection Service0.1.0OAS 3.1

default

GET /healthHealth

Report that the service is up and which model version it serves.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

```
{
  "additionalProp1": {}
}
```

POST /predictPredict

Score one transaction; the label is probability >= threshold and nothing else (BR-001).

Parameters

Try it out

No parameters

Request body

application/json

Example Value | Schema

```
{
  "Time": 0,
  "V1": 0,
  "V2": 0,
  "V3": 0,
  "V4": 0,
  "V5": 0,
  "V6": 0,
  "V7": 0,
  "V8": 0,
  "V9": 0,
  "V10": 0,
  "V11": 0,
  "V12": 0,
  "V13": 0,
  "V14": 0,
  "V15": 0,
  "V16": 0,
  "V17": 0,
  "V18": 0,
  "V19": 0,
  "V20": 0,
  "V21": 0,
  "V22": 0,
  "V23": 0,
  "V24": 0,
  "V25": 0,
  "V26": 0,
  "V27": 0,
  "V28": 0
}
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

```
{
  "request_id": "string",
  "probability": 0,
  "is_fraud": true,
  "threshold": 0,
  "model_version": "string",
  "latency_ms": 0
}
```

422	Validation Error	No links
-----	------------------	----------

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Schemas

HTTPValidationErrorExpand allobject

PredictionExpand allobject

TransactionExpand allobject

ValidationErrorExpand allobject

Screenshot 4 - Fraud test call. A fraud row from the test set sent to /predict. The probability is 1.0000, is_fraud is true and the answer came in 42 ms.

POST /predict - fraud transaction

```
--- fraud transaction (HTTP 200)
{
  "request_id": "51814alf-4185-49cb-86dd-a271d2d8bd6d",
  "probability": 0.9999999998115439,
  "is_fraud": true,
  "threshold": 0.7691996767092619,
  "model_version": "1",
  "latency_ms": 42.302
}
```

Screenshot 5 - Genuine test call. A genuine row from the test set. The probability is 0.028, below the threshold 0.7692, so is_fraud is false.

POST /predict - genuine transaction

```
--- genuine transaction (HTTP 200)
{
  "request_id": "84d86eb3-f540-4853-acfd-755e3984368d",
  "probability": 0.027732497926819904,
  "is_fraud": false,
  "threshold": 0.7691996767092619,
  "model_version": "1",
  "latency_ms": 2.469
}
```

Screenshot 6 - Prediction log. The prediction log after the test calls and the 100 latency requests: 102 rows, each with timestamp, request id, probability, label, latency and model version.

Prediction log (predictions_log.csv)

Rows logged: 102

	timestamp	request_id	probability	is_fraud	latency_ms	model_version
97	2026-10-08T01:10:48.230152+00:00	18979acb-7d4f-45cc-a613-96ba00b12067	0.027732	0	1.863	1
98	2026-10-08T01:10:48.234620+00:00	c2859a49-46f5-4923-959a-2545543ba792	0.027732	0	2.043	1
99	2026-10-08T01:10:48.239160+00:00	d5cc127b-9644-4555-baa8-08a09e0cece3	0.027732	0	2.508	1
100	2026-10-08T01:10:48.243046+00:00	41e02ad6-8d42-4eb4-8291-4077a14a7f28	0.027732	0	1.764	1
101	2026-10-08T01:10:48.246688+00:00	60300354-a249-4794-b1a0-c29eec5f0a28	0.027732	0	1.832	1

9. Conclusion

In this assignment we built a fraud detection system with a pipe-and-filter training pipeline, an MLflow registry and a FastAPI microservice that serves the champion model, with prediction logging and 31 automated tests. Against our goals: false-positive rate (1.08% against 2%) is met, latency (p95 5.9 ms against 200 ms) is met, and the test recall is 85.3% against the 90% target, so that goal is not met. The estimated fraud loss reduction is 82.4% against the 30% target (met). From this work we learned that with so few frauds in each split the recall on the test set can differ from the validation recall, that the model for serving should be chosen on validation results only, and that a registry alias keeps the service independent of the model version. If we had more time we would try more tuning on validation, cross-validation for the threshold, and more fraud examples.

Appendix A - Full Code

The complete code of every file is printed below. Repository:

<https://github.com/sanjaykumarpushadapu/SE-ML-Assessments> · Notebook: 13.ipynb

config.py

```
"""Single place for paths, constants and targets (BR-002, BR-011).

Every other file imports its settings from here, so a value such as the seed or the
target recall is never typed in two different places.
"""

from pathlib import Path

# Folder that contains this file (the project folder, i.e. where the notebook runs).
ROOT = Path(__file__).resolve().parent
SEED = 42 # fixed seed so every run gives the same split and the same model (BR-011)

# Where the data, outputs and the prediction log live.
DATA_PATH = ROOT / "data" / "creditcard.csv"
ARTIFACTS_DIR = ROOT / "artifacts" # metrics.json and sample requests are saved here
LOG_PATH = ROOT / "predictions_log.csv" # written by log_prediction() on every API call

# MLflow keeps its runs in a local SQLite file; model files go to the mlruns folder.
TRACKING_URI = f"sqlite:///({ROOT / 'mlflow.db'})"
ARTIFACT_ROOT = (ROOT / "mlruns").as_uri()
EXPERIMENT = "fraud-detection"
MODEL_NAME = "fraud-model" # name in the model registry
ALIAS = "champion" # the API only ever loads the model that has this alias (BR-009)

# Column names of the Kaggle dataset: Time, V1..V28 (anonymised), Amount; label is Class.
V_COLS = [f"V{i}" for i in range(1, 29)]
FEATURES = ["Time"] + V_COLS + ["Amount"]
SCALED_COLS = ["Time", "Amount"] # only these two are on very different scales, so only
they are scaled
TARGET = "Class" # 1 = fraud, 0 = genuine

# Measurable goals from the report (Section 2.2).
TARGET_RECALL = 0.90 # catch at least 90% of the frauds
MAX_FPR = 0.02 # wrongly flag less than 2% of genuine transactions
MIN_LOSS_REDUCTION = 0.30 # business goal: cut fraud loss by 30% (measured as the share of
fraud money caught)
MAX_P95_MS = 200.0 # 95% of API calls must answer within 200 ms
```

pipeline/ingest.py

```
"""Filter 1: read the raw CSV and check that the expected columns exist."""
from pathlib import Path

import pandas as pd

import config

def ingest(path: Path = config.DATA_PATH) -> pd.DataFrame:
    """Return the raw transactions; fail loudly if the file or columns are missing."""
    path = Path(path)
    # Stop with a clear message instead of continuing with made-up or empty data.
    if not path.exists():
        raise FileNotFoundError(f"Dataset not found at {path}. Upload creditcard.csv first.")
    df = pd.read_csv(path)
    # A wrong or damaged file should fail here, not later inside the model.
    missing = set(config.FEATURES + [config.TARGET]) - set(df.columns)
    if missing:
        raise ValueError(f"Dataset is missing columns: {sorted(missing)}")
    return df
```

pipeline/clean.py

```
"""Filter 2: remove duplicate rows and rows with missing values."""
import pandas as pd

import config

def clean(df: pd.DataFrame) -> pd.DataFrame:
    """Return a cleaned copy; the input frame is not modified."""
    needed = config.FEATURES + [config.TARGET]
    # drop_duplicates and dropna return new frames, so the caller's data stays unchanged.
    out = df.drop_duplicates().dropna(subset=needed).reset_index(drop=True)
    # The label must be exactly 0 or 1, otherwise the data is not what we expect.
    if not set(out[config.TARGET].unique()) <= {0, 1}:
        raise ValueError("Class column must contain only 0 and 1.")
    out[config.TARGET] = out[config.TARGET].astype(int)
    return out
```

pipeline/features.py

```
"""Filter 3: pick the model inputs (X) and the label (y).

Scaling is not done here. It is part of the model pipeline in train.py so that it
is fitted on the training set only (BR-003) and saved with the model (BR-004).
"""

import pandas as pd

import config

def build_features(df: pd.DataFrame) -> tuple[pd.DataFrame, pd.Series]:
    """Return the feature table and the label column."""
    # X holds the 30 input columns as floats; y holds the fraud label.
    return df[config.FEATURES].astype(float), df[config.TARGET]
```

pipeline/split.py

```
"""Filter 4: stratified 60/20/20 split into train, validation and test sets."""
import pandas as pd
from sklearn.model_selection import train_test_split

import config

def split(X: pd.DataFrame, y: pd.Series, seed: int = config.SEED) -> dict:
    """Return a dict with X_/y_ for train, val and test."""
    # Step 1: keep 20% aside as the test set. stratify=y keeps the same fraud share in each
    part,
    # which matters because fraud is only about 0.17% of the rows.
    X_rest, X_test, y_rest, y_test = train_test_split(
        X, y, test_size=0.20, stratify=y, random_state=seed)
    # Step 2: split the remaining 80% into 60% train and 20% validation (0.25 of 80% =
    20%).
    X_train, X_val, y_train, y_val = train_test_split(
        X_rest, y_rest, test_size=0.25, stratify=y_rest, random_state=seed)
    return {"X_train": X_train, "y_train": y_train, "X_val": X_val, "y_val": y_val,
            "X_test": X_test, "y_test": y_test}
```

pipeline/train.py

```
"""Filter 5: fit a model. The scaler and the classifier are one sklearn Pipeline."""
import pandas as pd
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import RobustScaler

import config

CANDIDATES = ("logistic_regression", "random_forest") # the two models we compare

def make_estimator(name: str, seed: int = config.SEED):
    """Return an unfitted classifier by name (this is the filter we can swap)."""
    if name == "logistic_regression":
        # class_weight="balanced" gives the rare fraud class a larger weight during
        training.
        return LogisticRegression(max_iter=1000, class_weight="balanced",
random_state=seed)
    if name == "random_forest":
        # max_depth and min_samples_leaf limit the tree size so training stays fast and
        # the forest does not memorise single transactions.
        return RandomForestClassifier(n_estimators=100, max_depth=12, min_samples_leaf=2,
class_weight="balanced_subsample", n_jobs=-1,
random_state=seed)
    raise ValueError(f"Unknown model name: {name}")

def train(X_train: pd.DataFrame, y_train: pd.Series, name: str) -> Pipeline:
    """Fit scaler and classifier on the training set only and return the pipeline."""
    # RobustScaler uses the median and quartiles, so a few very large Amount values
    # do not distort it. The other columns (V1..V28) are passed through unchanged.
    scale = ColumnTransformer([("scale", RobustScaler(), config.SCALED_COLS)],
remainder="passthrough")
    # Scaler + classifier in one object: the API loads this object, so serving applies
    # exactly the same scaling as training (BR-004).
    model = Pipeline([("prep", scale), ("clf", make_estimator(name))])
    model.fit(X_train, y_train) # the scaler is fitted on training data only (no leakage)
    return model
```

pipeline/evaluate.py

```
"""Filter 6: choose the threshold on validation data and compute the metrics."""
import numpy as np
import pandas as pd
from sklearn.metrics import (average_precision_score, confusion_matrix, f1_score,
                             precision_recall_curve, precision_score, recall_score,
                             roc_auc_score)

import config

def choose_threshold(y_true, proba, target_recall: float = config.TARGET_RECALL) -> float:
    """Highest threshold that still gives recall >= target on the given data."""
    # precision_recall_curve lists every possible threshold with its recall.
    # Recall goes down as the threshold goes up.
    _, recall, thresholds = precision_recall_curve(y_true, proba)
    # All thresholds that still reach the target recall.
    ok = np.where(recall[:-1] >= target_recall)[0]
    if len(ok) == 0:
        raise ValueError(f"Recall {target_recall} cannot be reached on this data.")
    # The last one is the highest such threshold, i.e. the fewest false alarms
    # while still meeting the recall target.
    return float(thresholds[ok[-1]])

def compute_metrics(y_true, proba, threshold: float) -> dict:
    """Recall, precision, F1, PR-AUC, ROC-AUC, FPR and the confusion matrix (BR-005)."""
    # The label is fraud when the probability reaches the threshold (BR-001).
    pred = (np.asarray(proba) >= threshold).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_true, pred, labels=[0, 1]).ravel()
    return {
        "threshold": float(threshold),
        "recall": float(recall_score(y_true, pred, zero_division=0)),          # frauds
        "precision": float(precision_score(y_true, pred, zero_division=0)),  # flagged that
        "f1": float(f1_score(y_true, pred, zero_division=0)),               are fraud
        "pr_auc": float(average_precision_score(y_true, proba)),            for rare classes
        "roc_auc": float(roc_auc_score(y_true, proba)),
        # false-positive rate = genuine transactions wrongly flagged / all genuine
        "fpr": float(fp / (fp + tn)) if (fp + tn) else 0.0,
        "tn": int(tn), "fp": int(fp), "fn": int(fn), "tp": int(tp),
    }

def targets_met(metrics: dict) -> dict:
    """Compare measured values with the targets; nothing is tuned here (BR-006)."""
    flags = {"recall_ok": metrics["recall"] >= config.TARGET_RECALL,
            "fpr_ok": metrics["fpr"] < config.MAX_FPR}
    if "loss_reduction" in metrics: # only the test report has the Amount-based loss
    figures
        flags["loss_ok"] = metrics["loss_reduction"] >= config.MIN_LOSS_REDUCTION
    return flags
```

```

def loss_metrics(y_true, proba, threshold: float, amount) -> dict:
    """Fraud loss measured in money (goal G4), assuming every flagged fraud is stopped.

    Loss = the Amount of the fraud transactions. Without the model every fraud goes
    through;
    with the model the flagged ones are blocked, so the loss reduction is the share of
    fraud money that is caught. The Amount of genuine transactions that are wrongly blocked
    is reported too, because that is the cost of the false alarms.
    """
    y = np.asarray(y_true)
    flagged = np.asarray(proba) >= threshold
    amt = np.asarray(amount, dtype=float)
    total = float(amt[y == 1].sum())
    caught = float(amt[(y == 1) & flagged].sum())
    return {"fraud_amount_total": total, "fraud_amount_caught": caught,
            "loss_reduction": caught / total if total else 0.0,
            "genuine_amount_blocked": float(amt[(y == 0) & flagged].sum())}

def evaluate_validation(model, X_val: pd.DataFrame, y_val: pd.Series) -> tuple[float,
dict]:
    """Pick the threshold on the validation set and report validation metrics."""
    proba = model.predict_proba(X_val)[:, 1] # column 1 = probability of fraud
    threshold = choose_threshold(y_val, proba)
    return threshold, compute_metrics(y_val, proba, threshold)

def evaluate_test(model, X_test: pd.DataFrame, y_test: pd.Series, threshold: float) ->
dict:
    """Final report on the test set with the threshold already fixed."""
    # The threshold comes from validation; the test set is never used to choose anything.
    proba = model.predict_proba(X_test)[:, 1]
    metrics = compute_metrics(y_test, proba, threshold)
    metrics.update(loss_metrics(y_test, proba, threshold, X_test["Amount"])) # fraud loss
    in money (G4)
    return metrics

```

pipeline/registry.py

```
"""MLflow tracking and model registry helpers (used by run_pipeline, not a filter)."""
import mlflow
from mlflow import MlflowClient

import config

def _setup() -> None:
    """Point MLflow at the local database and create the experiment once."""
    mlflow.set_tracking_uri(config.TRACKING_URI)
    # Create the experiment with a fixed artifact folder the first time only.
    if MlflowClient().get_experiment_by_name(config.EXPERIMENT) is None:
        mlflow.create_experiment(config.EXPERIMENT, artifact_location=config.ARTIFACT_ROOT)
    mlflow.set_experiment(config.EXPERIMENT)

def log_candidate(name: str, model, X_example, threshold: float, val_metrics: dict,
                 test_metrics: dict | None = None, register: bool = False):
    """Log one candidate as an MLflow run; register it when it is the winner."""
    _setup()
    with mlflow.start_run(run_name=name) as run:
        # Parameters: what was used. The API later reads "threshold" from here (BR-002).
        mlflow.log_param("model_name", name)
        mlflow.log_param("seed", config.SEED)
        mlflow.log_param("threshold", threshold)
        # Metrics: what came out (counts such as tn/fp are left out of the metric table).
        mlflow.log_metrics({f"val_{k}": v for k, v in val_metrics.items()
                           if k not in ("tn", "fp", "fn", "tp")})
        if test_metrics is not None: # only the winner is evaluated on the test set
            mlflow.log_metrics({f"test_{k}": v for k, v in test_metrics.items()
                               if k not in ("tn", "fp", "fn", "tp")})
        # cloudpickle is used because the default format refuses tree-based models.
        info = mlflow.sklearn.log_model(
            sk_model=model, name="model", input_example=X_example.head(2),
            serialization_format="cloudpickle",
            registered_model_name=config.MODEL_NAME if register else None)
    version = None
    if register:
        # Move the "champion" alias to the new version; the API loads whatever it points
        to.
        version = info.registered_model_version
        MlflowClient().set_registered_model_alias(config.MODEL_NAME, config.ALIAS, version)
    return run.info.run_id, version
```

pipeline/run_pipeline.py

```
"""Chains the filters: ingest, clean, features, split, train, evaluate (BR-010).

This file only connects the filters; the real work happens inside each filter.
"""

import json
import logging

import config
from pipeline.clean import clean
from pipeline.evaluate import evaluate_test, evaluate_validation, targets_met
from pipeline.features import build_features
from pipeline.ingest import ingest
from pipeline.split import split
from pipeline.train import CANDIDATES, train

log = logging.getLogger("pipeline")

def pick_winner(fitted: dict) -> str:
    """Candidate with the lowest validation false-positive rate; PR-AUC breaks ties.

    Every candidate already reaches the target recall on validation because its
    threshold was chosen for that, so the false-positive rate decides (goal G2).
    """
    return min(fitted, key=lambda n: (fitted[n][2]["fpr"], -fitted[n][2]["pr_auc"]))

def run_pipeline(data_path=config.DATA_PATH, candidates=CANDIDATES, track: bool = True) -> dict:
    """Run every stage, pick the candidate with the lowest validation false-positive rate,
    test it once."""
    # Each stage hands its output to the next one.
    raw = ingest(data_path)
    log.info("[1/6 ingest] rows=%d columns=%d", *raw.shape)

    df = clean(raw)
    log.info("[2/6 clean] rows=%d (removed %d), fraud=%d (%.3f%%)", len(df),
            len(raw) - len(df), df[config.TARGET].sum(), 100 * df[config.TARGET].mean())

    X, y = build_features(df)
    log.info("[3/6 features] X=%s", X.shape)

    parts = split(X, y)
    log.info("[4/6 split] train=%d val=%d test=%d", len(parts["X_train"]),
            len(parts["X_val"]), len(parts["X_test"]))

    # Train every candidate on the training set and judge it on the validation set.
    fitted = {}
    for name in candidates:
        model = train(parts["X_train"], parts["y_train"], name)
        threshold, val_m = evaluate_validation(model, parts["X_val"], parts["y_val"])
        fitted[name] = (model, threshold, val_m)
        log.info("[5/6 train] %-19s val recall=%.3f precision=%.3f pr_auc=%.4f
threshold=%.4f",
                name, val_m["recall"], val_m["precision"], val_m["pr_auc"], threshold)
```

```

# The winner is chosen from validation results only. Only now is the test set used,
once.
winner = pick_winner(fitted)
model, threshold, val_m = fitted[winner]
test_m = evaluate_test(model, parts["X_test"], parts["y_test"], threshold)
checks = targets_met(test_m) # real values against the targets, nothing is adjusted
log.info("[6/6 evaluate] winner=%s test recall=%.3f precision=%.3f f1=%.3f pr_auc=%.4f
fpr=%.4f",
        winner, test_m["recall"], test_m["precision"], test_m["f1"], test_m["pr_auc"],
        test_m["fpr"])
log.info("Targets on test set: recall>=%.2f -> %s, fpr<=%.2f -> %s",
config.TARGET_RECALL,
        checks["recall_ok"], config.MAX_FPR, checks["fpr_ok"])

result = {"winner": winner, "threshold": threshold, "val": val_m, "test": test_m,
        "targets_met": checks,
        "candidates": {n: {"threshold": t, "val": m} for n, (_, t, m) in
fitted.items()}}

if track: # track=False lets the tests run the pipeline without MLflow
    from pipeline.registry import log_candidate
    for name, (m, t, vm) in fitted.items():
        is_win = name == winner
        # Every candidate is logged for comparison; only the winner is registered.
        _, version = log_candidate(name, m, parts["X_train"], t, vm,
                                test_m if is_win else None, register=is_win)
        if is_win:
            result["model_version"] = version
            log.info("Registered %s version %s with alias '%s'", config.MODEL_NAME,
                    version, config.ALIAS)
    # Save results and one fraud + one genuine test row for the API test calls.
    config.ARTIFACTS_DIR.mkdir(exist_ok=True)
    (config.ARTIFACTS_DIR / "metrics.json").write_text(json.dumps(result, indent=2))
    samples = {
        "fraud": parts["X_test"][parts["y_test"] == 1].iloc[0].to_dict(),
        "genuine": parts["X_test"][parts["y_test"] == 0].iloc[0].to_dict()}
    (config.ARTIFACTS_DIR / "sample_requests.json").write_text(json.dumps(samples))
    return result

if __name__ == "__main__":
    logging.basicConfig(level=logging.INFO, format="%(message)s")
    run_pipeline()

```

api/prediction_log.py

```
"""Non-ML component: append one row per prediction to a CSV file (BR-008)."""
import csv
import threading
from datetime import datetime, timezone
from pathlib import Path

import config

FIELDS = ["timestamp", "request_id", "probability", "is_fraud", "latency_ms",
"model_version"]
_lock = threading.Lock() # two requests at the same time must not write into the file
together

def log_prediction(request_id: str, probability: float, is_fraud: bool, latency_ms: float,
                  model_version: str, path: Path | None = None) -> None:
    """Write one log row; an OSError is raised if the file cannot be written."""
    path = Path(path or config.LOG_PATH)
    # Only result data is logged. The transaction values themselves are not stored.
    row = {"timestamp": datetime.now(timezone.utc).isoformat(), "request_id": request_id,
          "probability": round(probability, 6), "is_fraud": int(is_fraud),
          "latency_ms": round(latency_ms, 3), "model_version": model_version}
    with _lock:
        new_file = not path.exists() or path.stat().st_size == 0 # write the header once
        with path.open("a", newline="") as f: # "a" = append, old rows are kept
            writer = csv.DictWriter(f, fieldnames=FIELDS)
            if new_file:
                writer.writeheader()
            writer.writerow(row)
```

api/main.py

```
"""Prediction microservice. It only loads the registered champion model; it never
trains."""
import logging
import time
import uuid
from contextlib import asynccontextmanager

import mlflow
import mlflow.sklearn
import pandas as pd
from fastapi import FastAPI, HTTPException, Request
from fastapi.exceptions import RequestValidationError
from fastapi.responses import JSONResponse
from mlflow import MlflowClient
from pydantic import BaseModel, ConfigDict, Field, create_model

import config
from api.prediction_log import log_prediction

log = logging.getLogger("api")

# Request schema: one required float for each of the 30 model inputs.
# allow_inf_nan=False rejects NaN and infinity; Amount must also be zero or more.
_fields = {c: (float, Field(..., allow_inf_nan=False)) for c in config.FEATURES}
_fields["Amount"] = (float, Field(..., ge=0, allow_inf_nan=False))
# extra="forbid" rejects unknown fields (for example a card number) with HTTP 422.
Transaction = create_model("Transaction", __config__=ConfigDict(extra="forbid"), **_fields)

class Prediction(BaseModel):
    """What the service sends back for one transaction."""
    request_id: str
    probability: float
    is_fraud: bool
    threshold: float
    model_version: str
    latency_ms: float

def load_champion():
    """Load models:/fraud-model@champion, its version and its threshold (BR-002, BR-
    009)."""
    mlflow.set_tracking_uri(config.TRACKING_URI)
    # Only the registered model with the champion alias is served, never a file or a run.
    model = mlflow.sklearn.load_model(f"models:{config.MODEL_NAME}@{config.ALIAS}")
    client = MlflowClient()
    mv = client.get_model_version_by_alias(config.MODEL_NAME, config.ALIAS)
    # The threshold was chosen during evaluation and stored as a run parameter,
    # so the service reads it instead of having its own copy.
    threshold = float(client.get_run(mv.run_id).data.params["threshold"])
    return model, str(mv.version), threshold

@asynccontextmanager
async def lifespan(app: FastAPI):
```

```

"""Runs once when the service starts: load the model into memory (keeps latency
low)."""
app.state.model, app.state.version, app.state.threshold = load_champion()
log.info("Loaded %s version %s, threshold %.4f", config.MODEL_NAME, app.state.version,
        app.state.threshold)
yield

app = FastAPI(title="Credit Card Fraud Detection Service", lifespan=lifespan)

@app.exception_handler(RequestValidationError)
async def validation_error_handler(request: Request, exc: RequestValidationError):
    """Return 422 without echoing the input (a NaN value cannot be serialised back)."""
    errors = [{"loc": list(e["loc"]), "msg": e["msg"], "type": e["type"]} for e in
exc.errors()]
    return JSONResponse(status_code=422, content={"detail": errors})

@app.get("/health")
def health() -> dict:
    """Report that the service is up and which model version it serves."""
    return {"status": "ok", "model": config.MODEL_NAME, "model_version": app.state.version,
            "threshold": app.state.threshold}

@app.post("/predict", response_model=Prediction)
def predict(tx: Transaction) -> Prediction:
    """Score one transaction; the label is probability >= threshold and nothing else (BR-
001)."""
    start = time.perf_counter()
    # Invalid requests never reach this point, FastAPI has already answered with 422.
    row = pd.DataFrame([tx.model_dump()], columns=config.FEATURES)
    # The saved model scales the input itself, so no separate preprocessing is done here
    (BR-004).
    probability = float(app.state.model.predict_proba(row)[0, 1])
    is_fraud = probability >= app.state.threshold
    latency_ms = (time.perf_counter() - start) * 1000
    request_id = str(uuid.uuid4()) # unique id so a log row can be matched to a call
    try:
        log_prediction(request_id, probability, is_fraud, latency_ms, app.state.version)
    except OSError as exc:
        # A failed log write is reported as an error, never ignored (BR-008).
        raise HTTPException(status_code=500, detail=f"Prediction log failed: {exc}") from
exc

return Prediction(request_id=request_id, probability=probability, is_fraud=is_fraud,
                  threshold=app.state.threshold, model_version=app.state.version,
                  latency_ms=round(latency_ms, 3))

```

tests/conftest.py

```
"""Shared fixtures. The tiny dataset here is synthetic and used for unit tests only."""
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import pytest

sys.path.insert(0, str(Path(__file__).resolve().parent.parent))

import config # noqa: E402

@pytest.fixture(scope="session")
def tiny_df() -> pd.DataFrame:
    """Small synthetic table (1000 rows, 50 frauds) used only by the tests, not the Kaggle
    data."""
    rng = np.random.default_rng(config.SEED)
    n, n_fraud = 1000, 50
    y = np.array([0] * (n - n_fraud) + [1] * n_fraud)
    X = rng.normal(0, 1, size=(n, 28)) + y[:, None] * 1.5
    df = pd.DataFrame(X, columns=config.V_COLS)
    df.insert(0, "Time", np.sort(rng.uniform(0, 172800, n)))
    df["Amount"] = rng.exponential(60, n)
    df["Class"] = y
    return df.sample(frac=1, random_state=config.SEED).reset_index(drop=True)

@pytest.fixture(scope="session")
def tiny_parts(tiny_df):
    """Train/validation/test parts of the tiny table, made with the real split filter."""
    from pipeline.features import build_features
    from pipeline.split import split
    X, y = build_features(tiny_df)
    return split(X, y)

@pytest.fixture(scope="session")
def tiny_model(tiny_parts):
    """A logistic regression pipeline fitted on the tiny training part with the real train
    filter."""
    from pipeline.train import train
    return train(tiny_parts["X_train"], tiny_parts["y_train"], "logistic_regression")

@pytest.fixture()
def valid_payload(tiny_df) -> dict:
    """One valid API request body taken from the tiny table."""
    row = tiny_df[config.FEATURES].iloc[0]
    return {k: float(v) for k, v in row.items()}
```

tests/test_pipeline.py

```
"""Tests for the pipeline filters and the rules BR-002 to BR-006, BR-010 and BR-011."""
import inspect
import json

import joblib
import numpy as np
import pandas as pd
import pytest

import config
from pipeline import clean as clean_mod, evaluate, features, ingest as ingest_mod, split as
split_mod
from pipeline import train as train_mod
from pipeline import run_pipeline as rp

# ---- BR-002 threshold chosen on validation -----
def test_BR002_threshold_meets_target_recall_on_validation(tiny_model, tiny_parts):
    """The chosen threshold must reach the target recall on the validation set."""
    thr, val_m = evaluate.evaluate_validation(tiny_model, tiny_parts["X_val"],
tiny_parts["y_val"])
    assert val_m["recall"] >= config.TARGET_RECALL
    assert val_m["threshold"] == thr

def test_BR002_unreachable_recall_raises():
    """If the target recall cannot be reached, choose_threshold must raise an error and not
guess."""
    with pytest.raises(ValueError):
        evaluate.choose_threshold([0, 1, 1], [0.9, 0.1, 0.2], target_recall=1.01)

def test_BR002_threshold_not_hard_coded_in_api():
    """The API must read the threshold from MLflow and must not contain its own fixed
value."""
    import api.main as api_main
    src = inspect.getsource(api_main)
    assert "params[\"threshold\"]" in src and ">= 0." not in src

# ---- BR-003 no leakage -----
def test_BR003_splits_are_disjoint_and_stratified(tiny_parts):
    """Train, validation and test must not share rows and must have the same fraud
share."""
    idx = [set(tiny_parts[f"X_{k}"].index) for k in ("train", "val", "test")]
    assert not (idx[0] & idx[1]) and not (idx[0] & idx[2]) and not (idx[1] & idx[2])
    rates = [tiny_parts[f"y_{k}"].mean() for k in ("train", "val", "test")]
    assert max(rates) - min(rates) < 0.01
    assert sum(len(i) for i in idx) == 1000

def test_BR003_scaler_fitted_on_train_only(tiny_model, tiny_parts, tiny_df):
    """The scaler must use the training median of Amount, not the median of all data (no
leakage)."""
    scaler = tiny_model.named_steps["prep"].named_transformers_["scale"]
```

```

train_median = tiny_parts["X_train"]["Amount"].median()
all_median = tiny_df["Amount"].median()
assert scaler.center_[1] == pytest.approx(train_median)
assert train_median != pytest.approx(all_median)

# ---- BR-004 same preprocessing in training and serving -----
def test_BR004_preprocessing_saved_inside_model(tiny_model, tiny_parts, tmp_path):
    """The scaler is saved inside the model, and a saved and reloaded model gives identical
    results."""
    assert "prep" in tiny_model.named_steps
    path = tmp_path / "m.pkl"
    joblib.dump(tiny_model, path)
    reloaded = joblib.load(path)
    X = tiny_parts["X_test"]
    np.testing.assert_allclose(reloaded.predict_proba(X), tiny_model.predict_proba(X))

# ---- BR-005 / BR-006 metrics and honest targets -----
def test_BR005_metrics_include_required_measures():
    """Metrics must include recall, precision, F1, PR-AUC and the confusion matrix, and not
    accuracy."""
    m = evaluate.compute_metrics([0, 0, 1, 1], [0.1, 0.6, 0.4, 0.9], 0.5)
    for key in ("recall", "precision", "f1", "pr_auc", "tn", "fp", "fn", "tp"):
        assert key in m
    assert "accuracy" not in m
    assert (m["tn"], m["fp"], m["fn"], m["tp"]) == (1, 1, 1, 1)

def test_BR006_targets_flag_is_false_when_target_missed():
    """When the targets are missed, the result must say so (False) instead of hiding it."""
    bad = evaluate.compute_metrics([0, 0, 1, 1], [0.1, 0.6, 0.4, 0.9], 0.5)
    flags = evaluate.targets_met(bad)
    assert flags["recall_ok"] is False and flags["fpr_ok"] is False

def test_BR006_test_set_not_used_for_threshold():
    """The threshold is chosen on validation data first; the test set is used only
    afterwards."""
    src = inspect.getsource(rp.run_pipeline)
    assert src.index("evaluate_validation") < src.index("evaluate_test")
    assert "choose_threshold" not in src

# ---- BR-010 pipe-and-filter -----
def test_BR010_filters_do_not_mutate_their_input(tiny_df):
    """Filters must return new data and leave the input table unchanged."""
    before = tiny_df.copy(deep=True)
    clean_mod.clean(tiny_df)
    features.build_features(tiny_df)
    pd.testing.assert_frame_equal(tiny_df, before)

def test_BR010_each_filter_is_a_separate_function():
    """Each filter is its own function in its own module."""
    for fn in (ingest_mod.ingest, clean_mod.clean, features.build_features,
split_mod.split,
train_mod.train, evaluate.evaluate_validation, evaluate.evaluate_test):

```

```

        assert callable(fn)
        assert len({f.__module__ for f in (ingest_mod.ingest, clean_mod.clean,
split_mod.split)}) == 3

def test_BR010_run_pipeline_chains_filters_in_order(monkeypatch, tiny_df, tmp_path):
    """run_pipeline must call the filters in the order ingest, clean, features, split,
train, evaluate."""
    calls = []

    def spy(name, fn):
        def wrapper(*a, **k):
            calls.append(name)
            return fn(*a, **k)
        return wrapper

    for name in ("ingest", "clean", "build_features", "split", "train",
"evaluate_validation",
                "evaluate_test"):
        monkeypatch.setattr(rp, name, spy(name, getattr(rp, name)))
    path = tmp_path / "creditcard.csv"
    tiny_df.to_csv(path, index=False)
    rp.run_pipeline(path, candidates=("logistic_regression",), track=False)
    order = ["ingest", "clean", "build_features", "split", "train", "evaluate_validation",
            "evaluate_test"]
    assert calls == order

def test_BR010_ingest_fails_loudly(tmp_path):
    """ingest must raise an error for a missing file and for a file with wrong columns."""
    with pytest.raises(FileNotFoundError):
        ingest_mod.ingest(tmp_path / "missing.csv")
    bad = tmp_path / "bad.csv"
    pd.DataFrame({"a": [1]}).to_csv(bad, index=False)
    with pytest.raises(ValueError):
        ingest_mod.ingest(bad)

# ---- BR-011 seeds -----
def test_BR011_seed_is_42_everywhere():
    """The configured seed and the models' random_state must be 42."""
    assert config.SEED == 42
    for name in train_mod.CANDIDATES:
        assert train_mod.make_estimator(name).random_state == 42

def test_BR011_same_seed_gives_same_split_and_results(tiny_df, tmp_path):
    """Running the pipeline twice must give exactly the same test results."""
    path = tmp_path / "c.csv"
    tiny_df.to_csv(path, index=False)
    a = rp.run_pipeline(path, candidates=("logistic_regression",), track=False)
    b = rp.run_pipeline(path, candidates=("logistic_regression",), track=False)
    assert json.dumps(a["test"], sort_keys=True) == json.dumps(b["test"], sort_keys=True)

def test_BR006_winner_is_lowest_validation_fpr_then_pr_auc():
    """The winner has the lowest validation false-positive rate; PR-AUC only breaks
ties."""

```

```
fitted = {"a": (None, 0.5, {"fpr": 0.04, "pr_auc": 0.9}),
          "b": (None, 0.5, {"fpr": 0.01, "pr_auc": 0.8}),
          "c": (None, 0.5, {"fpr": 0.01, "pr_auc": 0.7})}
assert rp.pick_winner(fitted) == "b"
```

```
def test_BR006_fraud_loss_reduction_is_share_of_fraud_money_caught():
    """Loss reduction = Amount of caught frauds / Amount of all frauds; genuine blocked
    money is reported too."""
    m = evaluate.loss_metrics([1, 1, 0, 0], [0.9, 0.1, 0.8, 0.2], 0.5, [100.0, 300.0, 50.0,
    70.0])
    assert m["fraud_amount_total"] == 400.0 and m["fraud_amount_caught"] == 100.0
    assert m["loss_reduction"] == 0.25 and m["genuine_amount_blocked"] == 50.0
```

tests/test_api.py

```
"""Tests for the prediction API: BR-001, BR-007, BR-008 and BR-009."""
import csv

import pytest
from fastapi.testclient import TestClient

import api.main as api_main
import config

@pytest.fixture()
def client(monkeypatch, tiny_model, tmp_path):
    """Test client for the API with a small model, threshold 0.5, model version 7 and a
    temporary log file."""
    monkeypatch.setattr(api_main, "load_champion", lambda: (tiny_model, "7", 0.5))
    monkeypatch.setattr(config, "LOG_PATH", tmp_path / "log.csv")
    with TestClient(api_main.app) as c:
        yield c

# ---- BR-001 -----
def test_BR001_label_is_probability_against_threshold(client, valid_payload):
    """The label must equal (probability >= threshold) and the probability must be between
    0 and 1."""
    r = client.post("/predict", json=valid_payload)
    assert r.status_code == 200
    body = r.json()
    assert 0.0 <= body["probability"] <= 1.0
    assert body["is_fraud"] == (body["probability"] >= body["threshold"])

def test_BR001_boundary_probability_equal_to_threshold_is_fraud(monkeypatch, tiny_model,
                                                                tmp_path, valid_payload):
    """A probability exactly equal to the threshold counts as fraud (boundary case)."""
    p = float(tiny_model.predict_proba(
        __import__("pandas").DataFrame([valid_payload], columns=config.FEATURES))[0, 1])
    monkeypatch.setattr(api_main, "load_champion", lambda: (tiny_model, "7", p))
    monkeypatch.setattr(config, "LOG_PATH", tmp_path / "log.csv")
    with TestClient(api_main.app) as c:
        assert c.post("/predict", json=valid_payload).json()["is_fraud"] is True

# ---- BR-007 -----
def test_BR007_missing_field_rejected(client, valid_payload):
    """A request with a missing feature must get HTTP 422."""
    del valid_payload["v5"]
    assert client.post("/predict", json=valid_payload).status_code == 422

def test_BR007_negative_amount_rejected(client, valid_payload):
    """A negative Amount must get HTTP 422."""
    valid_payload["Amount"] = -1.0
    assert client.post("/predict", json=valid_payload).status_code == 422
```

```

def test_BR007_wrong_type_rejected(client, valid_payload):
    """A text value where a number is expected must get HTTP 422."""
    valid_payload["V1"] = "abc"
    assert client.post("/predict", json=valid_payload).status_code == 422

def test_BR007_extra_field_rejected(client, valid_payload):
    """An unknown field (for example a card number) must get HTTP 422."""
    valid_payload["card_number"] = "4111111111111111"
    assert client.post("/predict", json=valid_payload).status_code == 422

@pytest.mark.parametrize("bad", ["NaN", "Infinity", "-Infinity"])
def test_BR007_nan_and_inf_rejected(client, valid_payload, bad):
    """NaN and infinity must get HTTP 422 and must not be logged."""
    parts = [f'"{k}": {bad if k == "V2" else v}' for k, v in valid_payload.items()]
    r = client.post("/predict", content="{" + ", ".join(parts) + "}",
                    headers={"content-type": "application/json"})
    assert r.status_code == 422
    assert not config.LOG_PATH.exists()

def test_BR007_invalid_input_is_not_logged(client, valid_payload):
    """A rejected request must not create a log row."""
    valid_payload["Amount"] = -5.0
    client.post("/predict", json=valid_payload)
    assert not config.LOG_PATH.exists()

# ---- BR-008 -----
def test_BR008_every_call_is_logged(client, valid_payload):
    """Each successful call writes one log row with the expected columns and no feature
    values."""
    for _ in range(3):
        assert client.post("/predict", json=valid_payload).status_code == 200
    rows = list(csv.DictReader(config.LOG_PATH.open()))
    assert len(rows) == 3
    assert set(rows[0]) == {"timestamp", "request_id", "probability", "is_fraud",
                           "latency_ms", "model_version"}
    assert rows[0]["model_version"] == "7"
    assert not any(k.startswith("V") for k in rows[0])

def test_BR008_log_failure_is_an_error_not_silent(monkeypatch, tiny_model, tmp_path,
valid_payload):
    """If the log cannot be written the API must answer HTTP 500, not continue silently."""
    monkeypatch.setattr(api_main, "load_champion", lambda: (tiny_model, "7", 0.5))
    monkeypatch.setattr(config, "LOG_PATH", tmp_path) # a directory cannot be opened as a
file
    with TestClient(api_main.app) as c:
        r = c.post("/predict", json=valid_payload)
        assert r.status_code == 500 and "log failed" in r.json()["detail"]

# ---- BR-009 -----
def test_BR009_loads_only_the_champion_alias(monkeypatch):
    """The API must load models:/fraud-model@champion and take the threshold from that
    model's run."""

```

```

seen = {}

class FakeMV:
    version = 3
    run_id = "abc"

class FakeRun:
    class data:
        params = {"threshold": "0.25"}

class FakeClient:
    def get_model_version_by_alias(self, name, alias):
        seen["alias"] = (name, alias)
        return FakeMV()

    def get_run(self, run_id):
        return FakeRun()

monkeypatch.setattr("mlflow.sklearn.load_model",
                    lambda uri: seen.setdefault("uri", uri) or "model")
monkeypatch.setattr(api_main, "MlflowClient", FakeClient)
_, version, threshold = api_main.load_champion()
assert seen["uri"] == "models:/fraud-model@champion"
assert seen["alias"] == ("fraud-model", "champion")
assert (version, threshold) == ("3", 0.25)

def test_BR009_response_carries_model_version(client, valid_payload):
    """Responses and /health must show the model version being served."""
    assert client.post("/predict", json=valid_payload).json()["model_version"] == "7"
    assert client.get("/health").json()["model_version"] == "7"

```

Notebook cells (setup, data check, EDA, run, test calls)

```
# ----- notebook cell 1 -----
# Cell: make sure the notebook runs inside its own project folder (all code files are
written here).
# If the kernel was started in a folder that no longer exists, stop with a clear message
instead of failing later.
import os
from pathlib import Path

try:
    os.getcwd()
except FileNotFoundError:
    if "__vsc_ipynb_file__" not in globals():
        raise RuntimeError("The kernel's working folder no longer exists. Restart the
kernel and run again.")
    if "__vsc_ipynb_file__" in globals():          # VS Code: use the folder that contains
13.ipynb
        os.chdir(Path(__vsc_ipynb_file__).parent)
print("Working folder:", os.getcwd())

# ----- notebook cell 2 -----
# Cell: install the extra packages (MLflow, FastAPI, uvicorn, pydantic, httpx, pytest,
requests).
%pip install -q mlflow fastapi uvicorn pydantic httpx pytest requests

# ----- notebook cell 3 -----
# Cell: create the project folders. The code files themselves are written by the
%%writefile cells below.
!mkdir -p data pipeline api tests artifacts docs/screenshots
!touch pipeline/__init__.py api/__init__.py

# ----- notebook cell 4 -----
# Cell: make sure data/creditcard.csv exists. In Colab it asks for an upload; elsewhere the
file must already be in data/.
# It stops with an error if the file is missing, so the notebook never continues with made-
up data.
from pathlib import Path
import importlib.util, os

if not Path("data/creditcard.csv").exists() and importlib.util.find_spec("google.colab"):
    from google.colab import files
    uploaded = files.upload()          # choose creditcard.csv
    for name in uploaded:
        os.replace(name, "data/creditcard.csv")

assert Path("data/creditcard.csv").exists(), "Put creditcard.csv in the data/ folder next
to this notebook"
print("Dataset ready:", Path("data/creditcard.csv").stat().st_size // 1_000_000, "MB")

# ----- notebook cell 5 -----
# Cell: first look at the raw data (shape, missing values, duplicates, class balance,
Amount and Time summary).
import pandas as pd
df_raw = pd.read_csv("data/creditcard.csv")
print("Shape:", df_raw.shape)
print("Missing values:", int(df_raw.isna().sum().sum()))
```

```

print("Duplicate rows:", int(df_raw.duplicated().sum()))
counts = df_raw["Class"].value_counts()
print(counts.to_string())
print(f"Fraud rate: {100 * df_raw['Class'].mean():.3f}%")
df_raw[["Time", "Amount"]].describe().T

# ----- notebook cell 6 -----
# Cell: plot the class balance and the Amount distribution to show how rare fraud is.
import matplotlib.pyplot as plt
fig, ax = plt.subplots(1, 2, figsize=(10, 3.5))
df_raw["Class"].value_counts().plot.bar(ax=ax[0], color=["#4C72B0", "#C44E52"])
ax[0].set_title("Class balance (0 = genuine, 1 = fraud)")
ax[0].set_xticklabels(["genuine", "fraud"], rotation=0)
df_raw["Amount"].clip(upper=500).plot.hist(bins=50, ax=ax[1], color="#4C72B0")
ax[1].set_title("Amount (clipped at 500)")
plt.tight_layout(); plt.show()

# ----- notebook cell 7 -----
# Cell: run the six-filter pipeline as a separate process. It trains both models, picks the
winner,
# tests it once and registers it in MLflow. Old MLflow data and the old prediction log are
removed first,
# so every full run starts clean and the registry always has version 1 as the champion.
# Warnings are filtered out of the display.
!rm -rf mlruns mlflow.db predictions_log.csv
!python -m pipeline.run_pipeline 2>&1 | grep -v -i warning

# ----- notebook cell 8 -----
# Cell: read artifacts/metrics.json and show validation results of both models next to the
test result of the winner.
import json, pandas as pd
res = json.load(open("artifacts/metrics.json"))
cols = ["threshold", "recall", "precision", "f1", "pr_auc", "roc_auc", "fpr"]
table = {f"{n} (validation)": c["val"] for n, c in res["candidates"].items()}
table[f"{res['winner']} (TEST, winner)"] = res["test"]
print("Winner (lowest validation FPR):", res["winner"], "| model version:",
res.get("model_version"))
display(pd.DataFrame(table).T[cols].round(4))
print("Targets on the test set:", res["targets_met"], "(recall >= 0.90, FPR < 0.02)")

# ----- notebook cell 9 -----
# Cell: draw the confusion matrix of the winner on the test set from the saved counts (tn,
fp, fn, tp).
import numpy as np, matplotlib.pyplot as plt
t = res["test"]
cm = np.array([[t["tn"], t["fp"]], [t["fn"], t["tp"]]])
fig, ax = plt.subplots(figsize=(4, 3.5))
ax.imshow(cm, cmap="Blues")
for (i, j), v in np.ndenumerate(cm):
    ax.text(j, i, f"{v}", ha="center", va="center", fontsize=13)
ax.set_xticks([0, 1], ["Predicted genuine", "Predicted fraud"])
ax.set_yticks([0, 1], ["Actual genuine", "Actual fraud"])
ax.set_title(f"Test confusion matrix ({res['winner']})")
plt.tight_layout(); plt.show()

# ----- notebook cell 10 -----
# Cell: show the MLflow runs (parameters and metrics) and the registered model with its
champion alias.

```

```

# This is done in a separate Python process so it always reads the current MLflow database
# (a kernel that stays open can keep an old, deleted database in memory after a re-run).
import subprocess, sys
query = ""
import mlflow, config
from mlflow import MlflowClient
mlflow.set_tracking_uri(config.TRACKING_URI)
runs = mlflow.search_runs(experiment_names=[config.EXPERIMENT])
keep = ["tags.mlflow.runName", "params.threshold", "metrics.val_recall",
"metrics.val_precision",
        "metrics.val_pr_auc", "metrics.test_recall", "metrics.test_precision",
"metrics.test_pr_auc"]
print(runs[[c for c in keep if c in runs.columns]].round(4).to_string())
client = MlflowClient()
for mv in client.search_model_versions(f"name='{config.MODEL_NAME}'"):
    print(f"Registered model: {mv.name} version: {mv.version} run: {mv.run_id}")
champ = client.get_model_version_by_alias(config.MODEL_NAME, config.ALIAS)
print(f"Alias '{config.ALIAS}' points to version {champ.version}")
"""

result = subprocess.run([sys.executable, "-W", "ignore", "-c", query], capture_output=True,
text=True)
print(result.stdout)
if result.returncode != 0:
    print(result.stderr[-1500:])

# ----- notebook cell 11 -----
# Cell: start the FastAPI service in the background and wait until /health answers (stops
with an error after 90 s).
import subprocess, time, requests
api = subprocess.Popen(["uvicorn", "api.main:app", "--port", "8000"],
                        stdout=open("api.log", "w"), stderr=subprocess.STDOUT)
for _ in range(90):
    try:
        health = requests.get("http://localhost:8000/health", timeout=2).json()
        break
    except requests.exceptions.ConnectionError:
        time.sleep(1)
else:
    raise RuntimeError("API did not start, see api.log:\n" + open("api.log").read())
print(health)

# ----- notebook cell 12 -----
# Cell: list the endpoints of the running service from its OpenAPI description.
spec = requests.get("http://localhost:8000/openapi.json").json()
for path, methods in spec["paths"].items():
    print(path, list(methods))

# ----- notebook cell 13 -----
# Cell: show the interactive /docs page inside Colab; outside Colab it prints the address
to open in a browser.
import importlib.util
if importlib.util.find_spec("google.colab"):
    from google.colab import output
    output.serve_kernel_port_as_iframe(8000, path="/docs", height=700)
else:
    print("Not running in Colab. Open http://localhost:8000/docs in your browser for the
screenshot.")

```

```

# ----- notebook cell 14 -----
# Cell: send one fraud row and one genuine row (saved by the pipeline from the test set) to
/predict.
import json
samples = json.load(open("artifacts/sample_requests.json"))
for kind, payload in samples.items():
    r = requests.post("http://localhost:8000/predict", json=payload)
    print(f"--- {kind} transaction (HTTP {r.status_code})")
    print(json.dumps(r.json(), indent=2))

# ----- notebook cell 15 -----
# Cell: send three invalid requests; each must be answered with HTTP 422 and must not be
logged.
bad_amount = dict(samples["genuine"], Amount=-5.0)
missing = {k: v for k, v in samples["genuine"].items() if k != "v5"}
extra = dict(samples["genuine"], card_number="4111111111111111")
for label, payload in [("negative Amount", bad_amount), ("missing v5", missing), ("unknown
field", extra)]:
    r = requests.post("http://localhost:8000/predict", json=payload)
    print(f"{label:16s} -> HTTP {r.status_code}", r.json()["detail"][0]["msg"])

# ----- notebook cell 16 -----
# Cell: send 100 requests and measure the response time to check the target p95 < 200 ms.
import config
import numpy as np, time
lat = []
for _ in range(100):
    t0 = time.perf_counter()
    requests.post("http://localhost:8000/predict", json=samples["genuine"])
    lat.append((time.perf_counter() - t0) * 1000)
p95 = float(np.percentile(lat, 95))
print(f"100 requests: median {np.median(lat):.1f} ms, p95 {p95:.1f} ms, max {max(lat):.1f}
ms")
print("p95 target met:", p95 < config.MAX_P95_MS)

# ----- notebook cell 17 -----
# Cell: show the prediction log; it should contain one row per successful call (the calls
above).
import config
log = pd.read_csv(config.LOG_PATH)
print("Rows logged:", len(log))
log.tail(5)

# ----- notebook cell 18 -----
# Cell: run all automated tests. The test names show which business rule (BR-xxx) each one
checks.
!python -m pytest -q -p no:warnings tests/ 2>&1 | tail -15

# ----- notebook cell 19 -----
# Cell: stop the background API process.
api.terminate(); api.wait(); print('API stopped')

```